Parameter efficient handling of acquisition artefacts to map tropical forests

Biomonweek 2026

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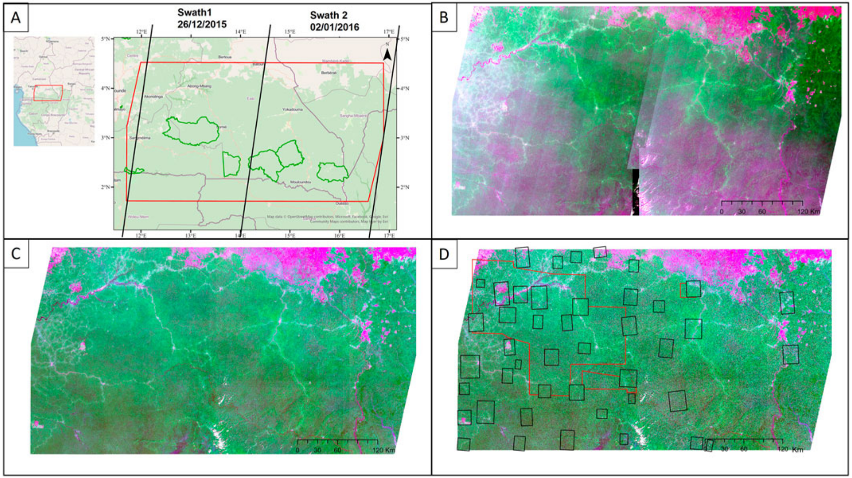
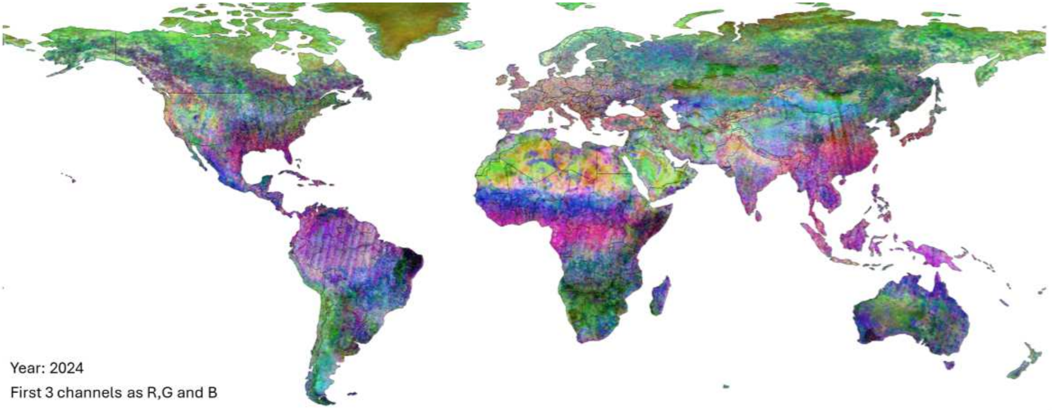
AMAPlab
CoForFunc
biodiversa+ European Biodiversity Partnership
Biodiversity Monitoring Week

Context: Mapping tropical forests is hard



Viennois et al., 2025

Context: Omnipotent foundation models ?



From Feng et al., 2025

Context: Omnipotent foundation models ?

- Not always at local scale
- Not always on niche tasks
- Not for all modalities
- Not always open source and open weights

Can we handle acquisition artefacts to map tropical forests ?

...

Can we handle acquisition artefacts to map tropical forests ?

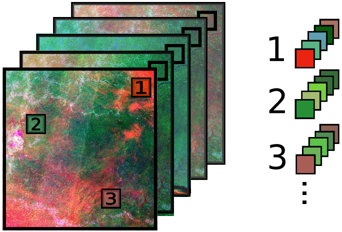
Without training a new expensive model

Constraints

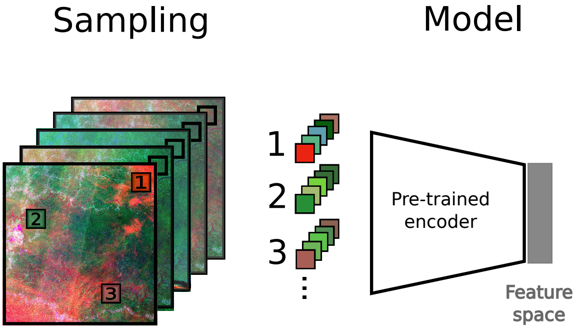
- Robust
- Versatile
- Actionnable without HPC
- Actionnable with limited labelled dataset

Methods : overview

Sampling

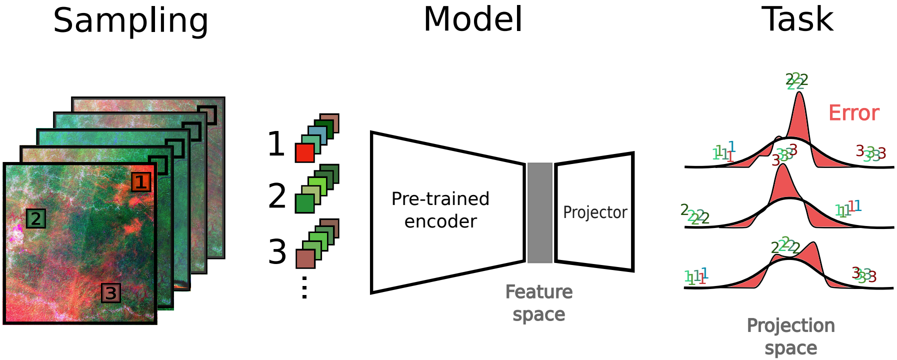


Methods : overview



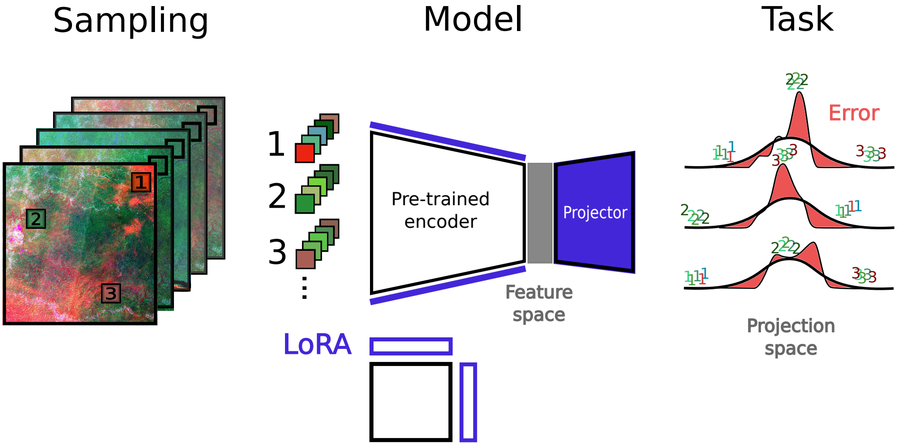
Encoders : DINOv3 and DOFA Siméoni et al., 2025; Xiong et al., 2024

Methods : overview



Based on LeJEPA Balestriero and LeCun, 2025

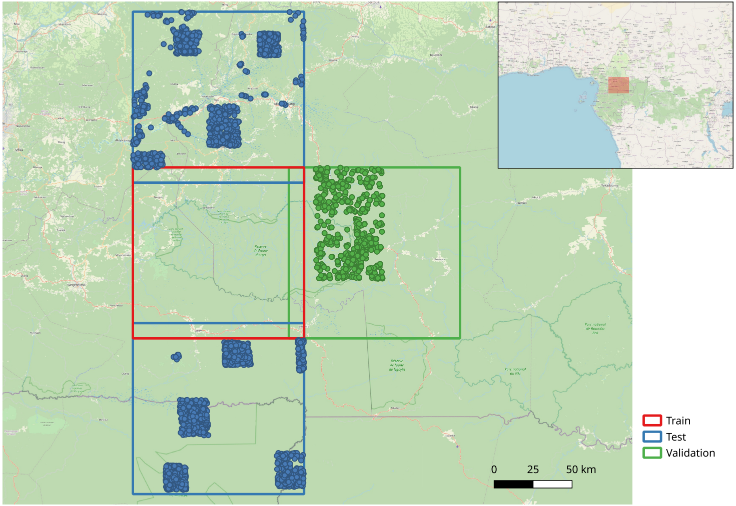
Methods : overview



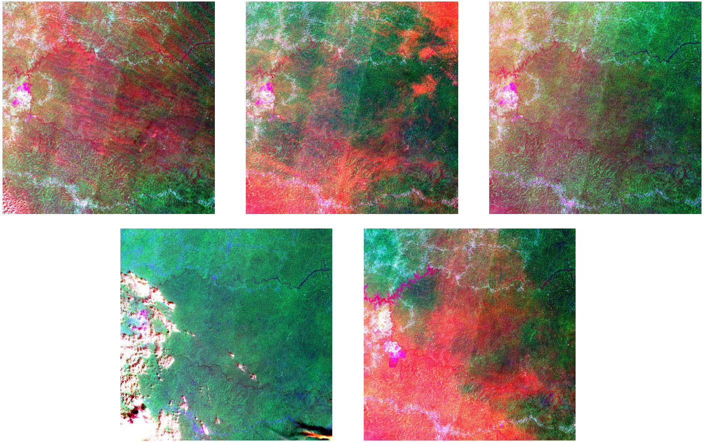
LoRA see Hu et al., 2022

Methods : datasets

- S2 tiles
- 655 labeled validation points
- 2103 labeled test points
- less than 15% clouds



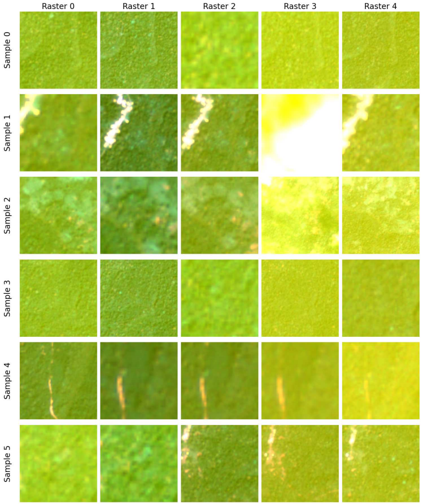
Methods : train dataset



RGB composition with bands 2, 8 and 11

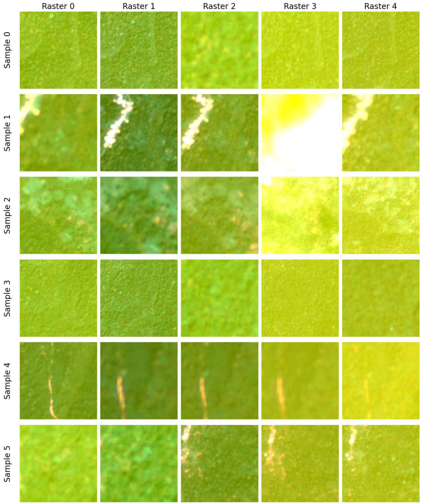
Methods : train sampling

- Random sampling on 5 dates



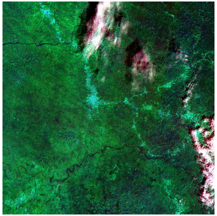
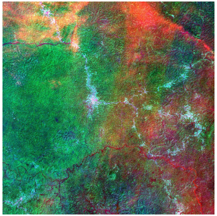
Methods : train sampling

- Random sampling on 5 dates
- Random zoom and crop (Between 30-150 pixels)

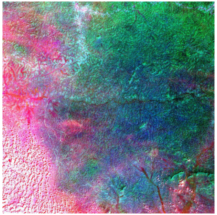
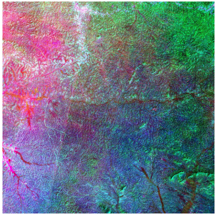
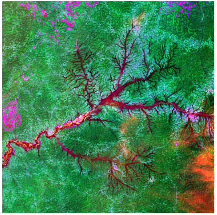
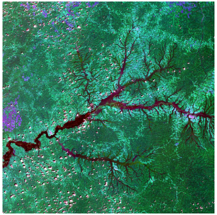


Methods : validation and test datasets

Validation :

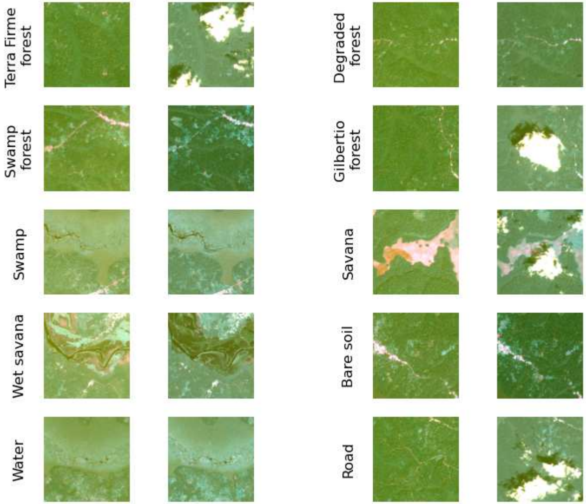



Test :

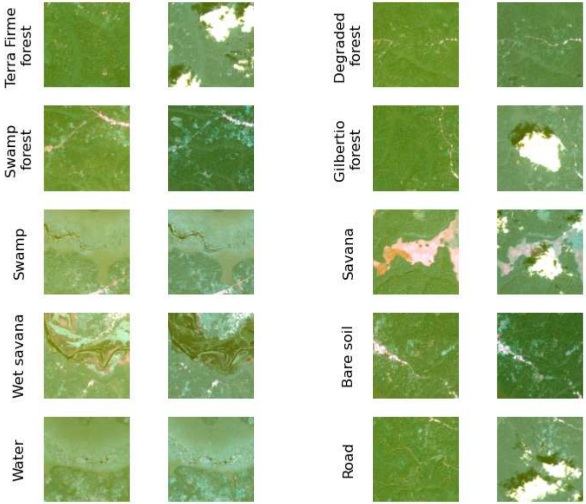
Methods : validation scheme

- 150 px centered on labeled point



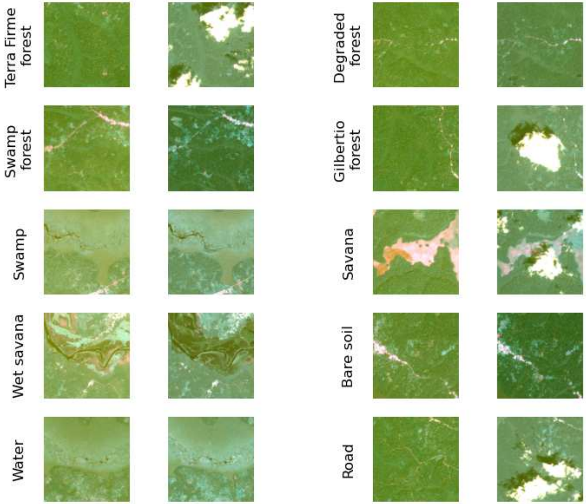
Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold



Methods : validation scheme

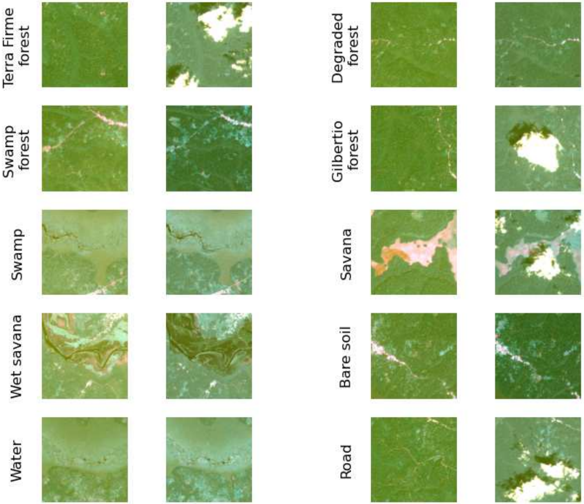
- 150 px centered on labeled point
- Geographical stratified 5-fold
- KNN fitted on model embedding of version A ...



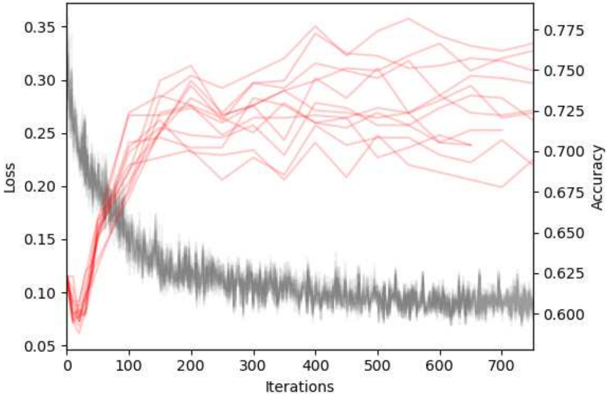
Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold
- KNN fitted on model embedding of version A …

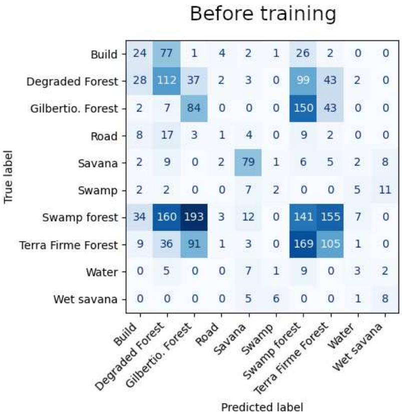
  … and tested on version B

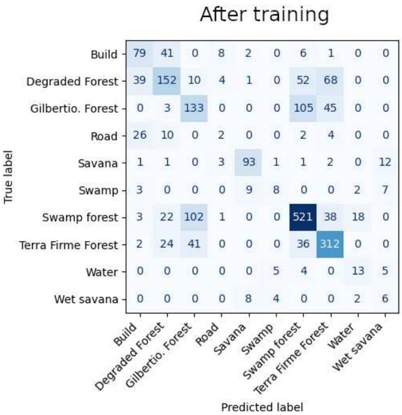


Results : training behaviour



Results : Confusion matrix

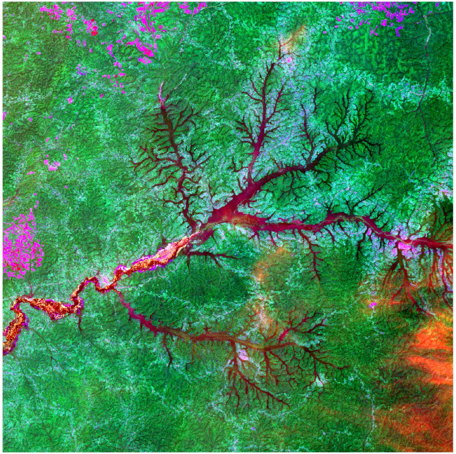


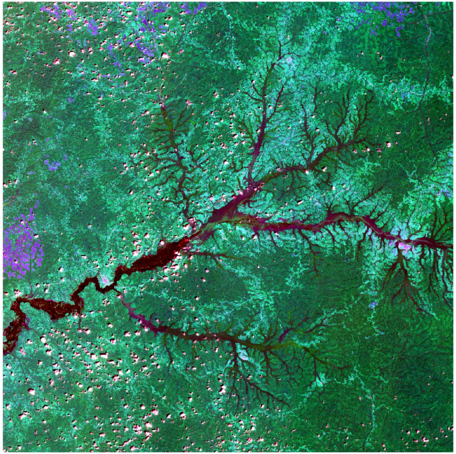


36% accuracy gain !

Visualisation

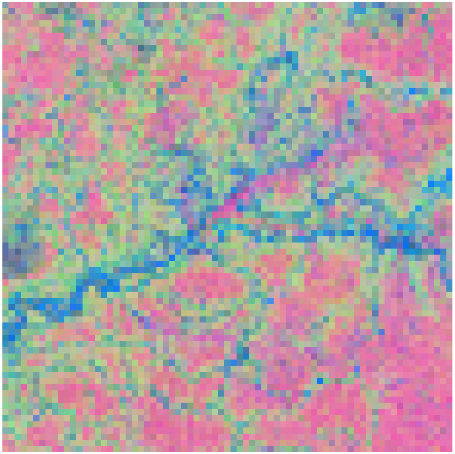
Original rasters

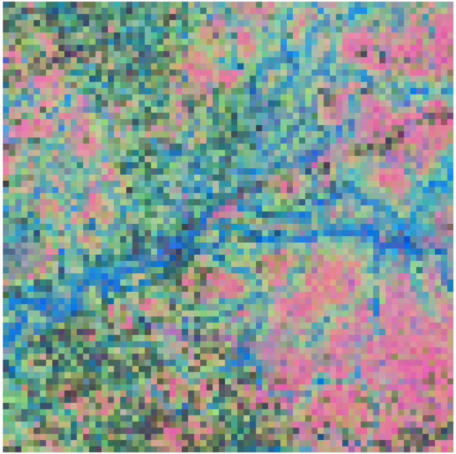




Visualisation : PCA

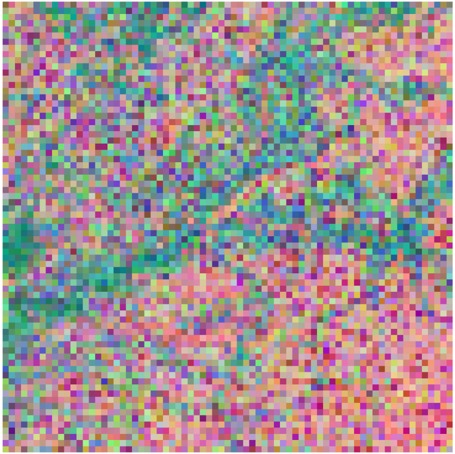
Before training

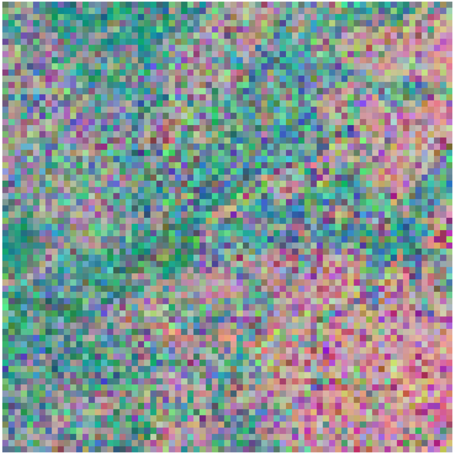




Visualisation : PCA

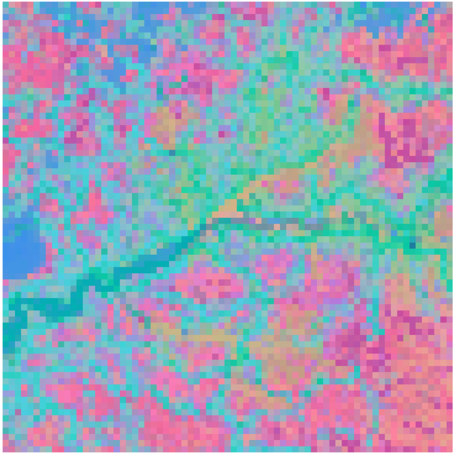
After training

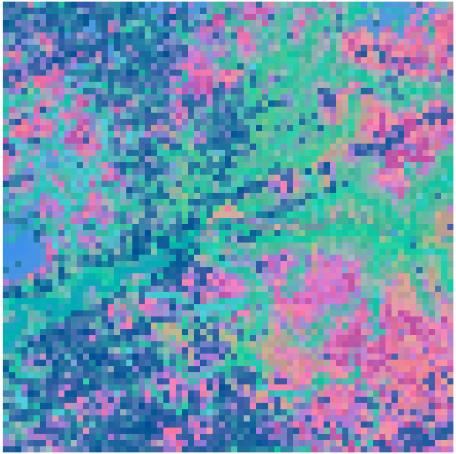




Visualisation : UMAP

Before training





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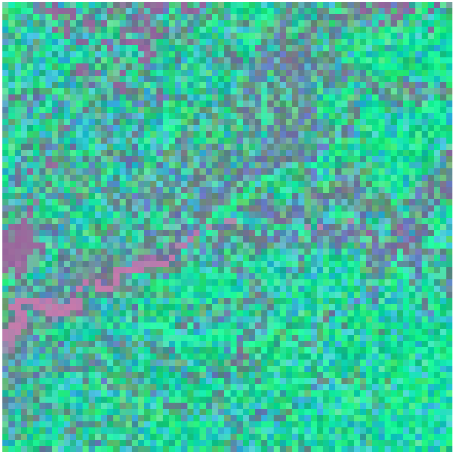
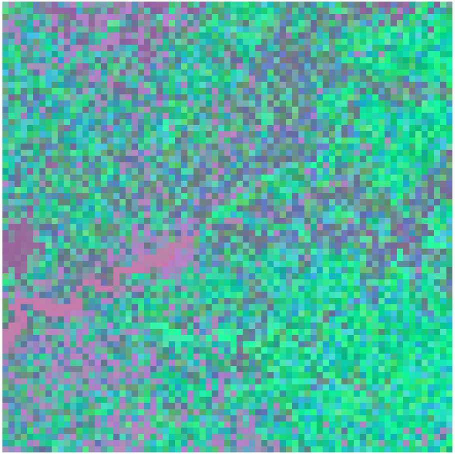
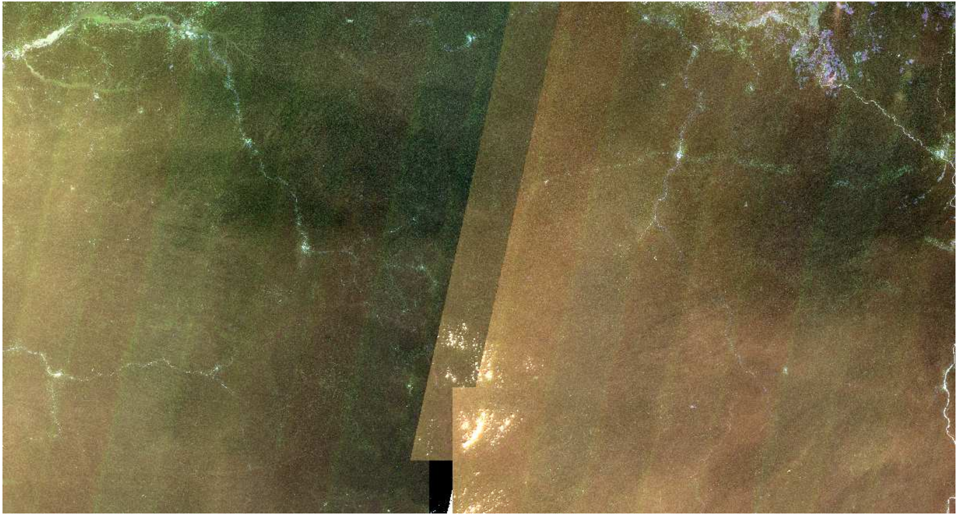
After training





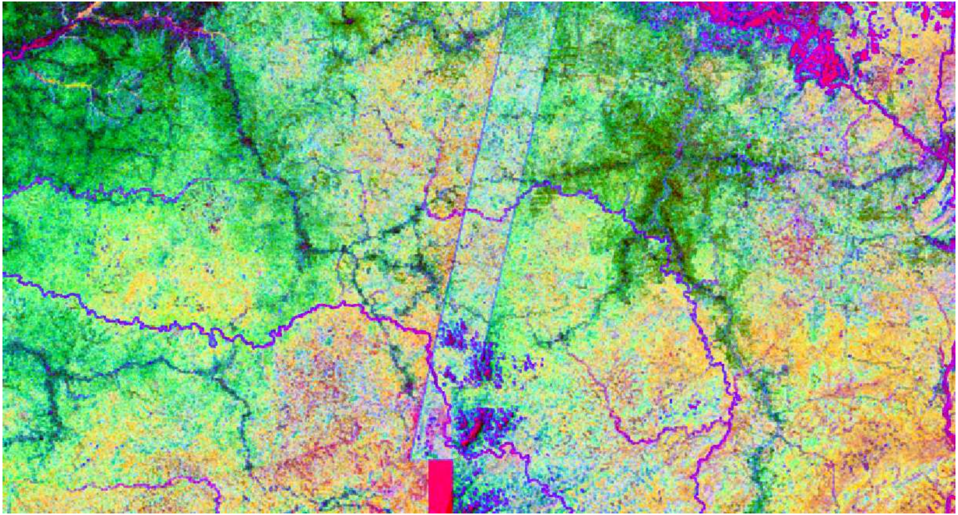
Scaling up

Original



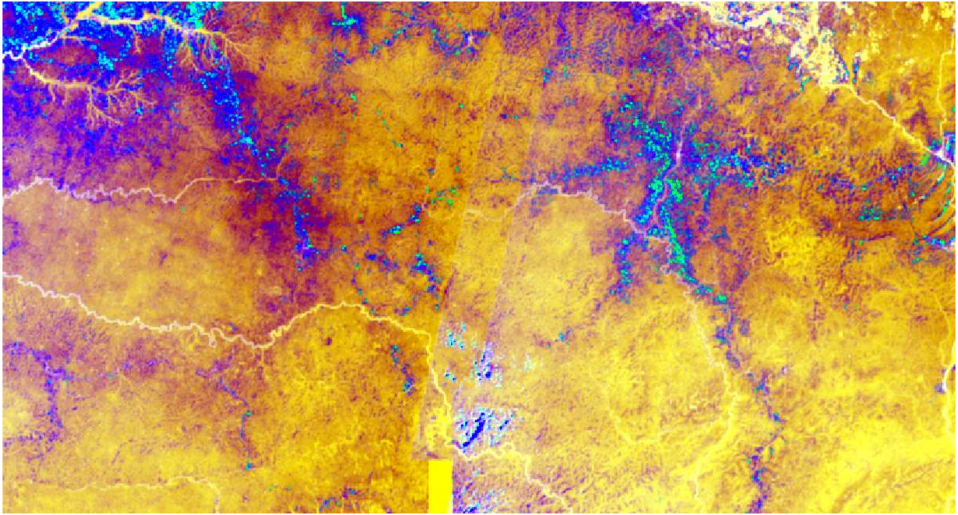
Scaling up

Before training



Scaling up

After training



Discussion

Not perfect yet but

- Only a couple hundred iterations (not epochs)

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  0.9 / 85.6M

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- Valid for any raster data

Discussion

Not perfect yet but

- Only a couple hundred iterations (not epochs)
- No labels needed
- Training only $\approx 1\%$ of the weights on a single GPU
  0.9 / 85.6M
- Valid for any raster data
- To map something **that does not vary between versions**

Thanks for your attention !

Any questions ?

Download this talk:

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References i

Balestriero, Randall and Yann LeCun (2025). **“Lejepa: Provable and scalable self-supervised learning without the heuristics”.** In: *arXiv preprint arXiv:2511.08544*.

Feng, Zhengpeng et al. (2025). **“Tessera: Temporal embeddings of surface spectra for earth representation and analysis”.** In: *arXiv preprint arXiv:2506.20380*.

Hu, Edward J et al. (2022). **“Lora: Low-rank adaptation of large language models.”.** In: *ICLR* 1.2, p. 3.

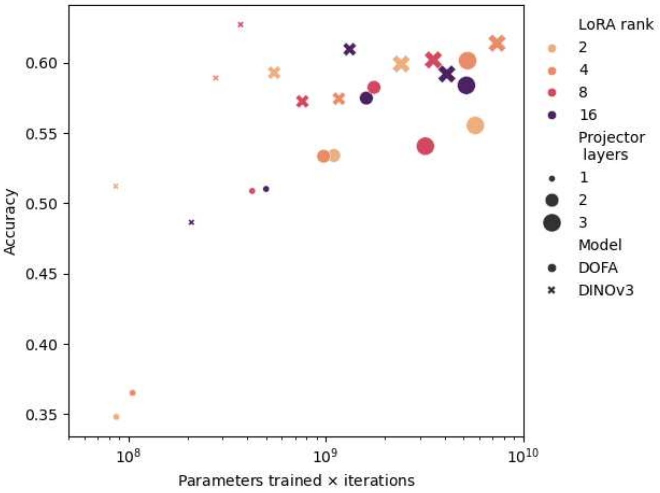
Siméoni, Oriane et al. (2025). **“Dinov3”.** In: *arXiv preprint arXiv:2508.10104*.

Viennois, Gaëlle et al. (2025). **“Sentinel-2 forest typology mapping in Central Africa: assessing deep learning and image preprocessing effects”.** In: *Frontiers in Remote Sensing* 6, p. 1682132.

References ii

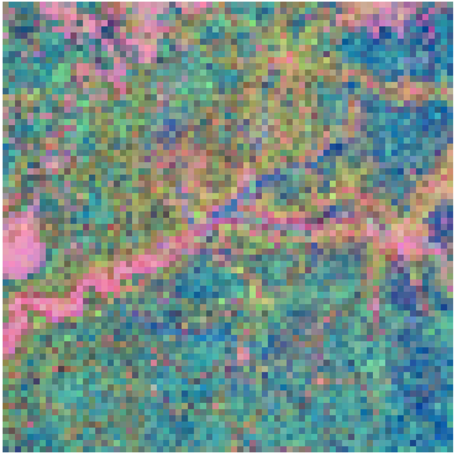
Xiong, Zhitong et al. (2024). **“Neural Plasticity-Inspired Foundation Model for Observing the Earth Crossing Modalities”.** In: *arXiv preprint arXiv:2403.15356*.

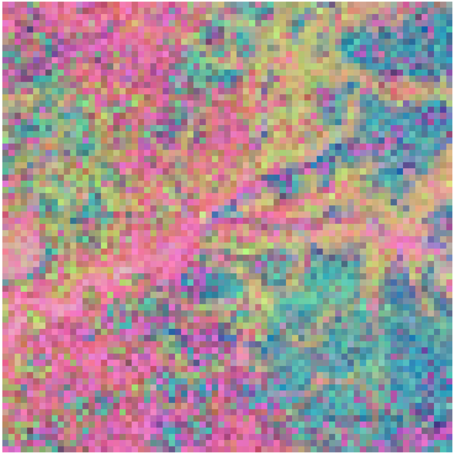
Results : Accuracy vs trained parameters



Visualisation DOFA PCA

Before training





Visualisation DOFA PCA

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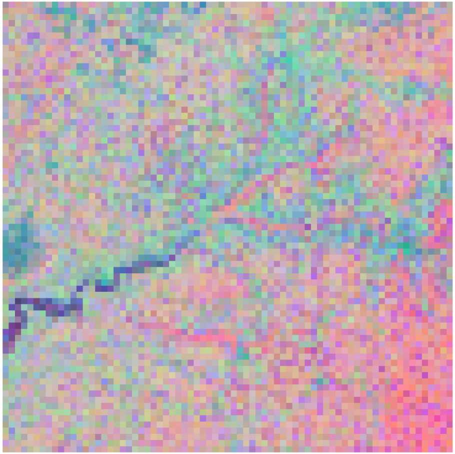

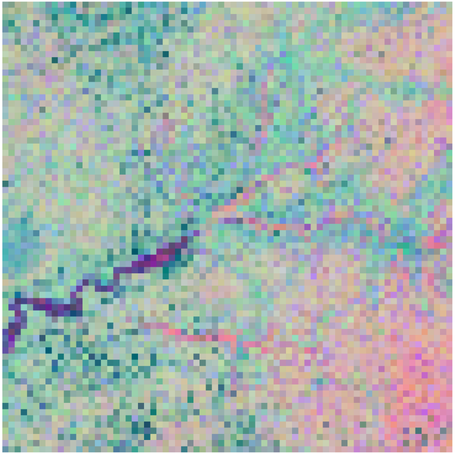

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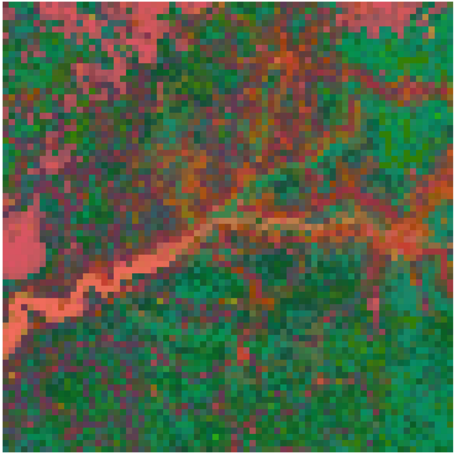

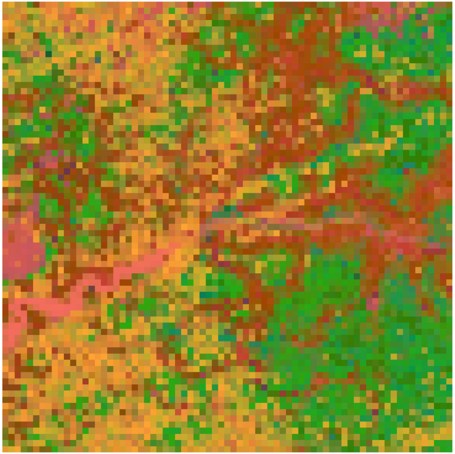

Visualisation DOFA UMAP

After training